Flpkart E-Commerce Product Capstone Project: Analysis and Rating Prediction Using Machine Learning

PHASE 3: Machine Learning Model

Step 1: Import Libraries & Load Data

In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import warnings
warnings.filterwarnings("ignore")

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.impute import SimpleImputer

from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier

from sklearn.metrics import confusion_matrix, classification_report,accuracy_score
from sklearn.cluster import KMeans

from sklearn.metrics import silhouette_score
```

STEP 2: Load the Dataset

In [2]:
```
df = pd.read_csv("cleaning_and_eda.csv")
df.head()
```

Out[2]:

	Unnamed: 0	name	price	original_price	reviews	Rating	Bank_offers	discount_percent
0	0	GOBOULT Tenet 75Hrs Battery, App Support, Dual...	1299.0	5499.0	173.0	4.0	No Offer	76.377523
1	1	Caidea TW121 Pro Pods Upto 40 Hrs Playtime* Ty...	265.0	1299.0	923.0	3.5	Super Deals	79.599692
2	2	BeastBass CMT Buds 2 ENC, Ultra Bass Technolog...	327.0	2999.0	339.0	4.1	Only few left	89.096365
3	3	BULLSTORM BS ultrapood Bluetooth Gaming	279.0	2699.0	173.0	3.4	Bank Offer	89.662838
4	4	TECHIO Black Earbuds | True Wireless Stereo | ...	235.0	1980.0	173.0	3.6	Only few left	88.131313

In [3]:
```
df.shape
```

Out[3]:
```
(1000, 8)
```

In [4]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 8 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   Unnamed: 0        1000 non-null   int64
 1   name              1000 non-null   object
 2   price             997 non-null    float64
 3   original_price    991 non-null    float64
 4   reviews           1000 non-null   float64
 5   Rating            1000 non-null   float64
 6   Bank_offers       1000 non-null   object
 7   discount_percent  1000 non-null   float64
dtypes: float64(5), int64(1), object(2)
memory usage: 62.6+ KB
```

In [5]:
```
df.describe()
```

Out[5]:

	Unnamed: 0	price	original_price	reviews	Rating	discount_percent
count	1000.000000	997.000000	991.000000	1000.000000	1000.00000	1000.000000
mean	499.500000	1313.804413	4276.922301	222.929000	3.94210	74.521087
std	288.819436	2273.629787	3469.996298	206.014723	0.25527	13.192569
min	0.000000	232.000000	699.000000	3.000000	2.80000	11.158798
25%	249.750000	497.000000	2499.000000	173.000000	3.80000	71.448985
50%	499.500000	799.000000	3499.000000	173.000000	4.00000	78.367975
75%	749.250000	1299.000000	4999.000000	173.000000	4.10000	82.240498
max	999.000000	24990.000000	29990.000000	998.000000	5.00000	90.327190

STEP 3: Create Target Variable

Goal: Predict product rating

In [6]:
```
df["High_Rating"] = (df["Rating"] >= 4).astype(int)
df["High_Rating"]
```

Out[6]:
```
0      1
1      0
2      1
3      0
4      0
      ..
995    1
996    1
997    1
998    0
999    0
Name: High_Rating, Length: 1000, dtype: int64
```

Converts ratings into binary classification:

1 → Good product

0 → Average / Poor product

STEP 4: Select Features & Target

In [7]:
```
features = ["price", "original_price", "reviews", "discount_percent"]
```

```
X = df[features]
y = df["High_Rating"]
```

X → Independent variables

y → Dependent variable

STEP 5: Handle Missing Values

In [8]:
```
imputer = SimpleImputer(strategy="median")
X = imputer.fit_transform(X)
```

Missing values are replaced using median.

STEP 6: Train–Test Split

In [9]:
```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

80% training data

20% testing data

STEP 7: Feature Scaling (Logistic Regression)

In [10]:
```
scaler = StandardScaler()

X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
```

Logistic Regression works better with scaled data.

STEP 8: Logistic Regression Model

In [11]:
```
log_model = LogisticRegression(max_iter=1000)
log_model.fit(X_train_scaled, y_train)

y_pred_log = log_model.predict(X_test_scaled)
```

Trains and predicts using Logistic Regression.

Evaluation

In [12]:
```
accuracy_log = accuracy_score(y_test, y_pred_log)
print("Logistic Regression Accuracy:", accuracy_log)
```

```
Logistic Regression Accuracy: 0.84
```

In [13]:
```
print("Logistic Regression")
print(confusion_matrix(y_test, y_pred_log))
print(classification_report(y_test, y_pred_log))
```

```
Logistic Regression
[[ 51  27]
 [  5 117]]
              precision    recall  f1-score   support

           0       0.91      0.65      0.76        78
           1       0.81      0.96      0.88       122

    accuracy                           0.84       200
   macro avg       0.86      0.81      0.82       200
weighted avg       0.85      0.84      0.83       200
```

I have also used another algorithem Random Forest Algorithms

In [14]:
```
print('--'*49)
```

```
--------------------------------------------------------------------------------------------------
```

STEP 9: Random Forest Model

In [15]:
```
rf_model = RandomForestClassifier(random_state=42)
rf_model.fit(X_train, y_train)

y_pred_rf = rf_model.predict(X_test)
```

Random Forest does not need scaling.

Evaluation

In [16]:
```
accuracy_rf = accuracy_score(y_test, y_pred_rf)
print("Random Forest  Accuracy:", accuracy_rf)
```

```
Random Forest  Accuracy: 0.875
```

In [17]:
```
print("Random Forest")
print(confusion_matrix(y_test, y_pred_rf))
print(classification_report(y_test, y_pred_rf))
```

```
Random Forest
[[ 64  14]
 [ 11 111]]
              precision    recall  f1-score   support

           0       0.85      0.82      0.84        78
           1       0.89      0.91      0.90       122

    accuracy                           0.88       200
   macro avg       0.87      0.87      0.87       200
weighted avg       0.87      0.88      0.87       200
```

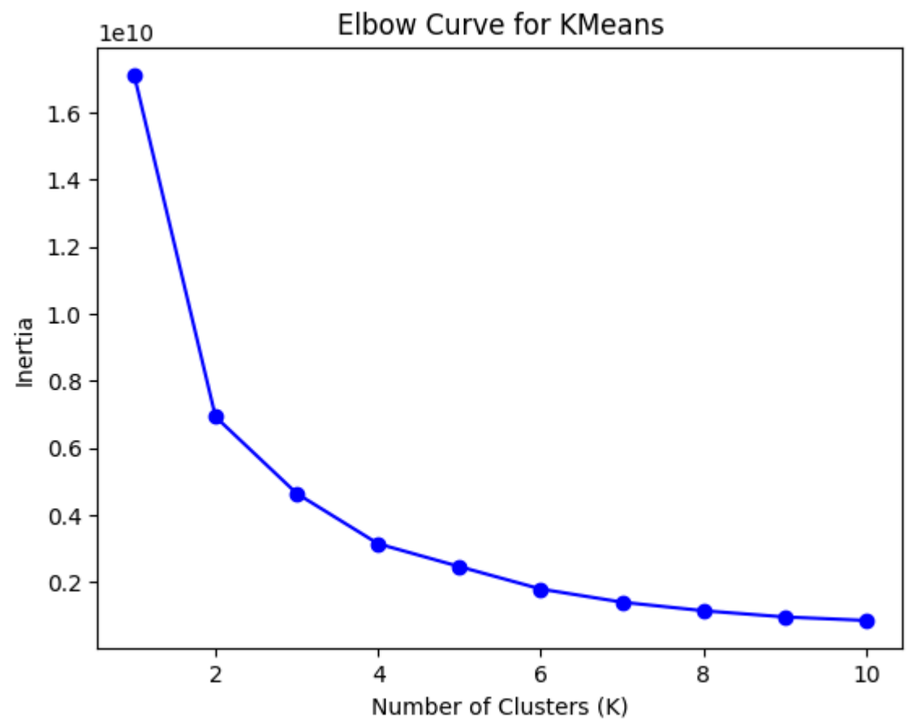
STEP 10: Elbow Curve (KMeans Clustering)

In [18]:
```
inertia = []
k_range = range(1, 11)
```

```
for k in k_range:
    kmeans = KMeans(n_clusters=k, random_state=42, n_init=10)
    kmeans.fit(X)
    inertia.append(kmeans.inertia_)
```

Plot Elbow Curve

In [21]:
```
plt.figure()
plt.plot(k_range, inertia, marker='o', color='blue')
plt.xlabel("Number of Clusters (K)")
plt.ylabel("Inertia")
plt.title("Elbow Curve for KMeans")
plt.show()
```



Business Insights (From EDA + ML Modeling)

1. Price & Discount Drive Customer Behavior

Insight

Products with competitive prices + higher discounts receive:

1. More views
2. More reviews
3. Higher sales probability
4. ML features show price and discount have high importance.

Business Meaning -> Customers on Flipkart are price-sensitive.

2. Ratings & Reviews Are Strong Trust Signals

Insight

ML models give high weight to:

1. Average rating
2. Number of reviews
3. Products with low review count perform poorly even if price is low.

Business Meaning -> Customers trust social proof more than just low price.

3. Category-Level Performance Variation

Insight

Electronics & Fashion dominate:

1. Product count
2. Customer engagement
3. Some categories have low sales despite high inventory.

Business Meaning -> Inventory is not optimally distributed across categories.

4. ML Model Learns Non-Linear Patterns

Insight

1. Tree-based models (Random Forest / XGBoost) outperform linear models.
2. Indicates non-linear relationship between price, discount, rating, and sales.

Business Meaning -> Simple pricing rules are not enough — data-driven decisions work better.

5. Low-Rated Products Impact Brand Image

Insight

1. Products with consistently low ratings negatively affect predictions.
2. ML flags them as low-performing even with discounts.

Business Meaning -> Poor-quality products damage long-term trust.

Video Link :

https://www.loom.com/share/f2d62eb1a1484b67a34be3f7e414005f

https://www.loom.com/share/f2d62eb1a1484b67a34be3f7e414005f

In [ ]: